Quantification: 548_hyp_HP_z14c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	29
interior lacuna count	21
frame-edge lacuna count	8

Interior lacunae only: mean, median and sample SD (n = 21)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2649.06	1397.5	2564.35	px^2
major_axis_length_px	lacuna shape	77.8706	66.1602	40.4921	px
minor_axis_length_px	lacuna shape	41.9268	34.8107	20.2484	px
aspect_ratio	lacuna shape	1.8804	2.009	0.4408	unitless
eccentricity	lacuna shape	0.7913	0.8673	0.1733	unitless
solidity	lacuna shape	0.7868	0.7974	0.0784	unitless
roots_count	headline	8.4762	7	5.8363	count
ring_length_r30_px	headline	273.667	218	174.766	px
ring_length_r60_px	reported	537.714	459	327.009	px
owned_length_px	ownership-dependent	818.698	613.044	754.835	px
edge_count	ownership-dependent	43.0952	22	41.5607	count
mean_edge_length_px	ownership-dependent	19.9202	18.8043	4.4041	px
perimeter_px	normalised	243.63	192.989	179.953	px
ring_area_r30_px2	normalised	7659.57	6390	4406.65	px^2
ring_area_r60_px2	normalised	16437.3	14742	8615.3	px^2
in_frame_fraction_r30	normalised	0.9755	1	0.0607	unitless
in_frame_fraction_r60	normalised	0.9494	1	0.1003	unitless
ring_density_r30	normalised	0.0364452	0.0342391	0.0100246	px^-1
ring_density_r60	normalised	0.032825	0.0323261	0.00709844	px^-1
roots_per_100px_perimeter	normalised	4.0803	3.4832	1.9422	per 100 px
ring_attached_length_r30_px	network v2	240.571	167	170.068	px
ring_attached_length_r60_px	network v2	404.095	305	302.782	px
ring_length_w_r30_px	network v2	312.585	260.392	199.469	px
ring_length_w_r60_px	network v2	614.027	520.375	375.049	px
sholl_crossings_r10	network v2	6.3333	5	4.0042	count
sholl_crossings_r20	network v2	7.0952	7	4.7948	count
sholl_crossings_r30	network v2	7.2857	6	5.1589	count
loop_count	loops, report only	2.4286	2	2.7124	count
loop_edge_share	loops, report only	0.168	0.1429	0.1389	share
ring_width_mean_r30_px	width	7.7359	7.5689	0.6818	px

Per field
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measure	value	units
analysed_area_px2	978753	px^2
total_skeleton_length_px	27424	px
canalicular_length_density_per_px	0.0280193	px^-1
junction_count	593	count
junction_density_per_px2	0.00060587	px^-2
lacuna_count	29	count

lacunae_per_px2	2.963e-05	px^-2
skeleton_component_count	385	count
mean_component_length_px	71.2312	px
median_component_length_px	20	px
field_length_density_w_per_px	0.0318795	px^-1
field_density_without_flagged_per_px	0.0280193	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	248	px
unattached_component_count	14	count
canalicular_length_density_attached_per_px	0.0280193	px^-1
field_length_density_with_unattached_per_px	0.0282727	px^-1
loop_count	70	count
loop_edge_share	0.1705	share
edge_count	1548	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8593	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	11.1507	px
width_pixels_used	27424	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2147	91.4001	39.1166	2.3366	0.9038	0.7373	0.5027	31.0005	542	True
2	2407	118.141	31.4268	3.7592	0.964	0.7314	0.0789	45.1836	688.173	True
3	704	43.5023	24.5984	1.7685	0.8248	0.8253	0.3934	22.6634	726.641	True
4	3074	97.1472	47.5979	2.041	0.8717	0.7951	0.7084	40.3224	458.502	False
5	4688	114.763	56.7615	2.0218	0.8691	0.8409	-0.5667	73.3957	882.268	False
6	477	43.1425	19.7158	2.1882	0.8895	0.6903	0.7532	160.568	566.931	False
7	996	48.8421	30.1974	1.6174	0.786	0.83	-0.3922	177.571	118.507	False
8	1527	72.7312	32.1897	2.2595	0.8967	0.7689	1.1031	215.125	119.538	False
9	791	45.4603	27.379	1.6604	0.7983	0.777	-1.1935	276.899	15.5398	True
10	885	49.967	25.4783	1.9612	0.8602	0.8157	0.1758	321.858	27.7831	False
11	865	60.9988	23.7735	2.5658	0.9209	0.67	-0.6206	306.805	864.442	False
12	9764	174.84	79.9051	2.1881	0.8895	0.756	-1.0035	341.592	370.654	False
13	546	27.4648	26.7738	1.0258	0.2229	0.901	-0.4745	337.135	652.104	False
14	5590	100.499	74.9477	1.3409	0.6662	0.8931	0.3458	390.83	706.214	False
15	772	53.4057	21.7698	2.4532	0.9131	0.7546	-1.3439	371.161	29.9093	False
16	3806	88.412	63.5037	1.3922	0.6958	0.7996	0.5855	456.034	447.32	False
17	1368	68.8013	32.6693	2.106	0.8801	0.7621	0.1327	437.326	13.0395	True
18	818	48.7035	23.1908	2.1001	0.8794	0.8503	0.5594	490.792	650.099	False
19	583	30.9882	25.7454	1.2036	0.5565	0.8807	-0.6814	486.81	163.602	False
20	6173	146.612	73.4479	1.9961	0.8655	0.639	0.4437	581.134	750.702	False
21	1268	55.1715	37.4316	1.4739	0.7346	0.792	0.9741	664.745	233.587	False
22	1408	71.6194	28.9459	2.4743	0.9147	0.8167	1.0982	723.629	987.69	True
23	3872	116.659	53.0473	2.1992	0.8906	0.6702	-1.541	793.571	878.724	False
24	3821	95.6808	52.75	1.8139	0.8343	0.9065	-1.1603	884.495	983.684	True

25	1979	71.3216	39.2045	1.8192	0.8354	0.8154	-1.307	916.707	906.786	False
26	8745	138.777	89.9937	1.5421	0.7612	0.8115	1.2047	976.542	708.753	True
27								861.622	179.927	False
28								977.516	631.554	False
29								63.9025	760.855	False

Network per lacuna

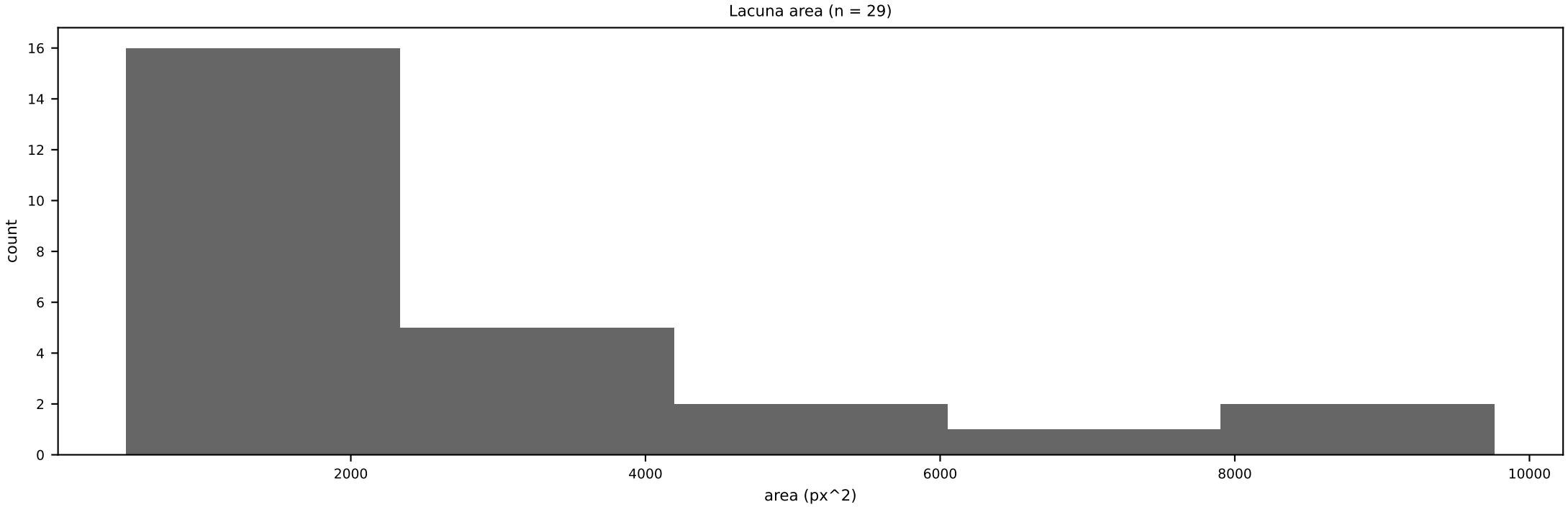
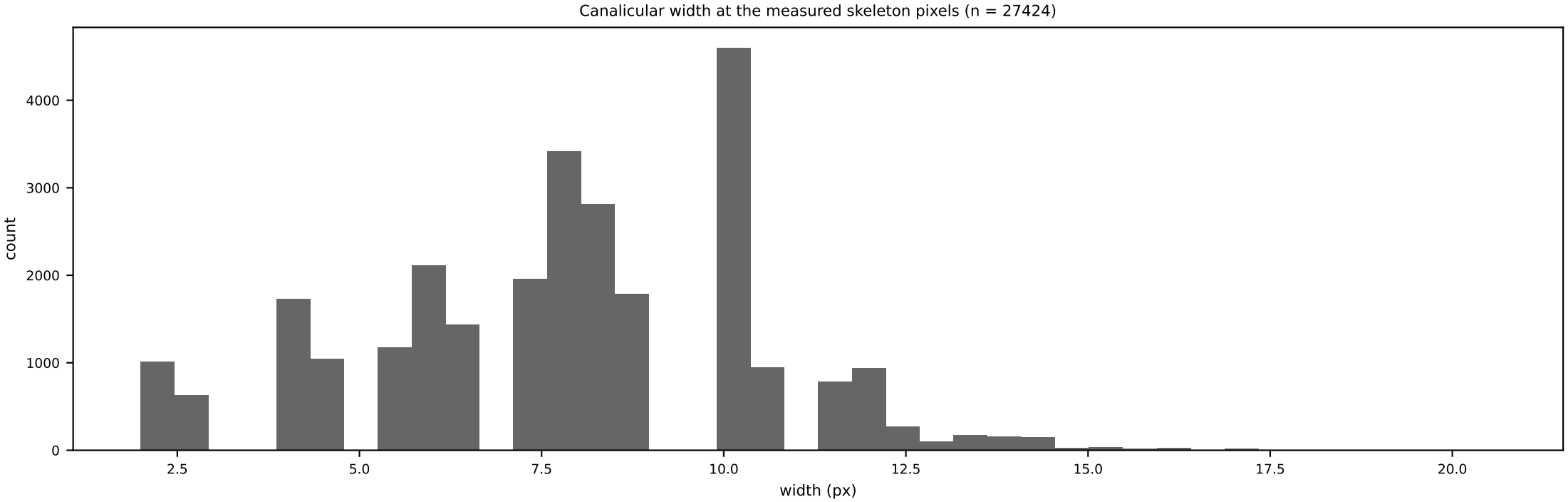
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	182	235	563.286	21	26.8232	6.9664
2	11	300	555	480.961	29	16.5848	8.0307
3	4	75	103	66.2843	6	11.0474	4.5702
4	10	241	459	954.909	57	16.7528	7.7676
5	7	288	527	736.566	22	33.4803	8.7175
6	9	286	545	360.149	22	16.3704	6.9746
7	4	123	248	551.132	27	20.4123	8.0682
8	6	172	421	613.044	22	27.8656	7.5689
9	3	151	227	63.1838	2	31.5919	5.7979
10	6	70	94	75.9411	4	18.9853	6.6883
11	12	321	689	1428.11	80	17.8514	7.3687
12	24	778	1522	3351.14	182	18.4129	7.3534
13	4	191	456	1019.29	42	24.2689	9.0428
14	7	332	599	1107.69	68	16.2896	9.0851
15	5	168	247	225.652	12	18.8043	7.2723
16	10	353	751	911.453	50	18.2291	7.1225
17	1	59	133	83.8406	4	20.9602	7.0119
18	4	218	399	443.274	21	21.1083	7.4342
19	7	184	566	283.451	21	13.4977	7.9669
20	22	607	1128	1972.56	99	19.9248	7.5777
21	7	274	570	813.784	44	18.4951	7.2242
22	8	311	574	1431.67	71	20.1643	7.755
23	17	525	861	1278.74	78	16.3942	7.831
24	6	201	375	404.09	23	17.5691	8.0704
25	5	175	358	319.208	19	16.8004	8.5769
26	7	353	714	499.345	27	18.4943	6.8839
27	3	96	325	132.024	6	22.0041	7.462
28	5	131	220	232.995	11	21.1814	7.0099
29	4	214	307	381.534	18	21.1963	8.3408

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	314.877	6899	10287	0.7247	0.6734	0.0263806	0.0228444	2.2231
2	330.693	7326	14532	0.7916	0.7301	0.04095	0.0381916	3.3263
3	132.368	2205	2673	0.7504	0.6702	0.0340136	0.0385335	3.0219
4	289.563	8243	15166	0.836	0.748	0.0292369	0.0302651	3.4535
5	351.161	12196	24989	0.9903	0.8855	0.0236143	0.0210893	1.9934
6	136.853	6220	14728	1	1	0.0459807	0.0370043	6.5764
7	192.989	4006	9474	1	1	0.0307039	0.0261769	2.0727
8	221.38	6390	14742	1	1	0.0269171	0.0285579	2.7103
9	143.74	3169	5742	0.6661	0.6142	0.0476491	0.0395333	2.0871
10	148.61	2877	4028	0.8558	0.7127	0.0243309	0.0233366	4.0374
11	184.652	7467	20230	1	1	0.0429891	0.0340583	6.4987
12	750.649	18042	35989	1	1	0.0431216	0.0422907	3.1972
13	100.711	3823	9996	1	1	0.0499608	0.0456182	3.9718
14	352.919	10336	19147	1	1	0.0321207	0.0312843	1.9835
15	186.959	3412	6249	0.8041	0.7079	0.049238	0.0395263	2.6744

16	328.233	11363	23232	1	1	0.0310657	0.0323261	3.0466
17	193.539	4187	8710	0.6366	0.5949	0.0140912	0.0152698	0.5167
18	137.196	6367	12630	1	1	0.0342391	0.0315915	2.9155
19	104.569	5596	16799	1	1	0.0328806	0.0336925	6.6942
20	631.595	16663	33826	1	1	0.036428	0.0333471	3.4832
21	199.865	7779	21047	1	1	0.035223	0.0270822	3.5024
22	197.48	6605	13178	0.8037	0.7245	0.0470855	0.0435574	4.051
23	394.747	12775	22894	1	1	0.0410959	0.0376081	4.3066
24	281.38	6112	10015	0.7455	0.6918	0.0328861	0.0374438	2.1323
25	221.865	6537	13263	1	1	0.0267707	0.0269924	2.2536
26	525.044	10840	21452	0.7555	0.698	0.0325646	0.0332836	1.3332
27	54.5269	4382	14411	1	1	0.0219078	0.0225522	5.5019
28	53.1127	2511	5429	1	0.8984	0.0521705	0.0405231	9.4139
29	74.0833	3866	6914	1	0.9847	0.0553544	0.0444027	5.3993

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	173	226	197.397	261.752	4	3	2
2	209	287	349.534	645.227	7	6	8
3	71	71	86.4264	118.154	3	1	0
4	188	295	275.451	520.375	5	7	4
5	278	430	336.363	609.784	9	9	10
6	243	305	329.078	630.399	3	6	7
7	123	182	138.154	271.409	4	4	2
8	152	309	194.61	474.747	4	5	6
9	70	70	173.782	259.622	1	2	4
10	70	70	78.0122	105.154	4	0	0
11	291	580	378.99	797.838	6	10	6
12	719	1338	884.767	1740.22	18	17	20
13	101	270	214.853	515.576	3	7	5
14	331	555	380.049	676.872	8	9	5
15	160	224	189.024	283.007	5	2	3
16	257	377	394.191	861.394	11	8	17
17	39	39	61.3848	143.154	1	2	3
18	126	283	260.392	473.943	3	7	5
19	167	329	206.51	643.529	4	4	9
20	580	904	697.612	1291.34	11	16	14
21	255	480	318.907	643.872	6	8	7
22	306	518	353.25	648.63	5	8	9
23	499	821	591.002	981.436	13	17	14
24	168	295	232.551	427.333	6	6	7
25	153	207	194.782	394.978	5	4	6
26	298	449	394.12	793.784	8	9	6
27	64	89	114.468	377.233	2	2	5
28	128	157	143.083	243.995	5	2	1
29	167	281	243.995	357.463	4	5	7

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